

ARNAV SANGHAVI

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[LinkedIn](#) | [Portfolio](#)

➤ SUMMARY

AAA Technical Designer who shipped Gears of War: E-Day. Connects Design and Programming teams using Unreal Engine 5, Blueprints, C++, and AI systems. With special Interest in gameplay mechanics, Level Design tools, and feature optimization.

➤ CORE SKILLS

- **Languages & Scripting:** C++, C#, Blueprint Visual Scripting
- **Engine:** Unreal Engine 5, Unity
- **Systems:** AI Systems (Behavior Trees, EQS, State Trees, Smartobjects), Gameplay Systems
- **Optimization & Tools:** Profiling & Optimization, World Partition, Custom Editor Tools, Maya
- **Collaboration & Workflow:** Perforce, Git, Jira, Technical Documentation, Mentorship

➤ PROFESSIONAL EXPERIENCE

- **Technical Gameplay Designer | The Coalition Studio (Microsoft)** Feb 2024 - Present
 - **Refactored smartobjects and NPC behaviors**, replacing complicated workflows with scalable architecture to eliminate designer bottlenecks and speed up content creation.
 - **Designed and optimized enemy spawn closets**, preventing frame drops and ensuring predictable combat pacing in encounters.
 - **Converted core gameplay systems from Blueprint to C++** (smartobjects, narrative events, spawn closets), eliminating CPU performance spikes and execution overhead.
 - **Built custom editor tools that automated repetitive setups and improved asset visibility**, reducing manual overhead and designer search time
 - **Acted as the key technical escalation point for Level and Narrative teams**, diagnosing and fixing critical script errors and systemic blockers in live builds.
 - **Restructured Narrative Graphs and barks implementation with Narrative Designers**, solving data disorganization to ensure clean, scalable graph executions.
 - **Scaled team capability and script quality** by authoring technical documentation, running bi-weekly info sessions & setup reviews, and enforcing strict scripting standards.
- **Game Programmer 1 | BLINK Media Inc.** Oct 2022 - Feb 2024
 - Programmed **multiplayer gameplay, horse movement, and player controls**.
 - Created **custom tools for game designers** to speed up content creation.
 - Migrated the **audio architecture to FMOD**, replacing the legacy manager with an event-based system.

➤ RELEASED GAMES

- **Gears of War: E-Day** | Technical Gameplay Designer
Unreal Engine, C++, NPC behavior systems, systemic gameplay, AI-driven features
- **FEI Equestriad World Tour** | Game Programmer
Gameplay systems, multiplayer mechanics, player movement, Unity, FMOD

➤ EDUCATION

- **Advanced Game Technology Diploma** 2020 - 2022
Kwantlen Polytechnic University
- **Bachelor of Science in Information Technology** 2017 - 2019
Mumbai University